



NichePhone-S is a card-sized phone

From FutureModel, the NichePhone-S is a minimal smartphone with Android 4.2, Bluetooth support and wireless charging. <http://dgit.in/NichePhnS>



Google assistant can now identify songs

No longer exclusive to the Pixel 2, Google has rolled out song identification to all phones with the Assistant. <http://dgit.in/GoogleSong>

- Feature extraction: Relevant parameters are extracted.
- Matching: Extracted parameters are matched against pre-existing data-set.

Going into the details of every established approach would require a treatise if not a book itself on facial recognition. Methods like eigenfaces, linear discriminant analysis, elastic bunch graph matching using the Fisherface algorithm, the hidden Markov model, the multilinear subspace learning using tensor representation, and the neuronal motivated dynamic link matching have all been tried and tested across years and have been used in various combination to make up the market leading face recognition APIs of today.

In general there are a couple of approaches that are identifiable across these algorithms. While statistical analysis gets rid of all the visuals and combines the parameters directly, in the form of raw numbers, it is the neural network approach that is currently state of the art. That, combined with 3D modelling techniques is what makes today's face recognition systems what they are.

3D FACE RECOGNITION

With 3-dimensional face recognition, things get truly state of the art. This method, overall, is way more accurate than its 2D counterparts and rivals fingerprint systems in accuracy. Some of the advantages of using 3D modelling for face recognition are pretty evident. For instance, it doesn't have to rely as much on environmental factors like lighting, angle, head orientation etc.



One of the USPs of Apple's iPhone X is the face recognition system, FaceID

Additionally, most 3D scanners of today also capture visual data along with structural information. This allows most 3D recognition techniques to incorporate the traditional

2D algorithms as well, increasing the overall accuracy. In more recent examples, Apple's iPhone X features a facial recognition system that reportedly utilises both 2D and 3D sensors to provide more accurate response.

APIS AND SDKS

From revolutionising payments to bringing in a completely new dimension to personalised ads, there are a multitude of reasons why the world would be crazy for the state-of-the-



Amazon Rekognition is one of the most comprehensive offerings that has the added advantage of being well-supported on AWS

art in facial recognition tech. But not everybody possesses the technical knowhow or the tools to delve into it at an algorithmic level. At that point, established APIs and SDKs come to the rescue. There are quite a few options available in the market:

- Amazon Rekognition
- Google Vision API
- Microsoft Face API
- IBM Watson
- Cognitec
- NEC
- Affectiva
- OpenCV

The details between these APIs mainly differ on specific features. For instance, Amazon Rekognition only charges you for the number of images you process and the face metadata you store, but video formats are not supported, which, on the other hand, is supported by OpenCV. But OpenCV, which is only available as an SDK,

doesn't recognise age, gender and emotions, which is detected by Microsoft Face API and quite a few other alternatives as well. The final decision, on which API to go for, rests absolutely on the use case as well as volume of data it will be used for.

FUTURE OF FACE RECOGNITION

Enough has been said about the privacy concerns surrounding face recognition and its ubiquitous use. While that is a discussion for another

day, it is inevitable to explore some of the debatable plans when you look at the future of face recognition. For instance, higher resolution cameras as well as thermal cameras in combination are being developed to identify health markers on your face (you can already detect pulse using a webcam). Think of the future of facial recognition as being more encompassing in the metrics it is able to collect. Soon, you'll have something not too different from a fingerprint being generated from the lines on your face (ear recognition is already a quite popular talk point in biometric authentication). Taking it one step further from today's capabilities, facial recognition could incorporate muscle movement analysis (the way you smile, down to the muscles involved) bringing behavioural analysis into everyday consumer gadgets. It's all about how precise things can get from here on. 